

TARGET-90

JKBOSE HUMANITIES CLASS- 11TH

MOTION IN STRAIGHT LINE

One Shot

PHYSICS

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Topics to be Covered



A Kinematics

B Motion & Rest

C Distance & Displacement

D Speed, Velocity and Acceleration



$$\textcircled{3} \quad S = ut + \frac{1}{2}at^2$$

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$$S = \frac{1}{2}(v+u) \textcircled{t}$$

$$a = \frac{(v-u)}{t}$$

$$t = \frac{(v-u)}{a}$$

$$S = \frac{1}{2}(v+u) \frac{(v-u)}{a}$$

$$\begin{aligned} &\rightarrow (a+b)(a-b) \\ &= a^2 - b^2 \end{aligned}$$

$$2as = (v+u)(v-u)$$

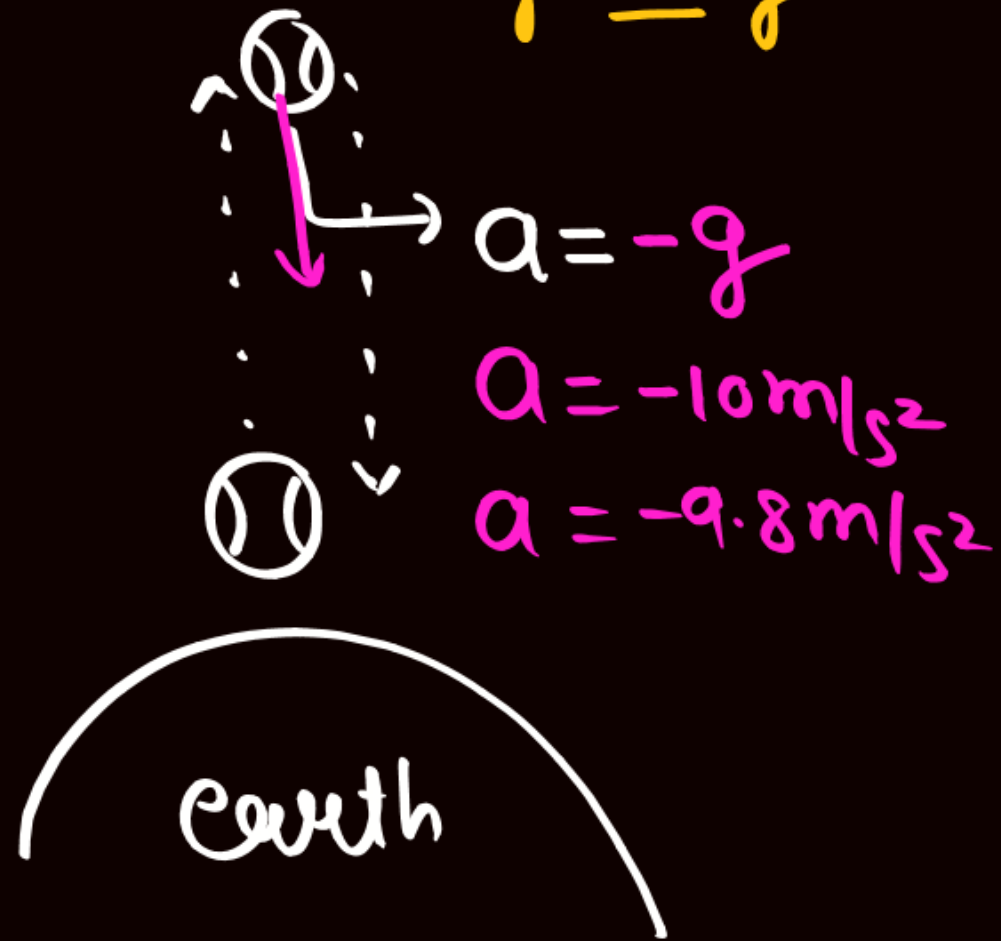
$$2as = v^2 - u^2$$

$$u^2 + 2as = v^2$$

$$v^2 = u^2 + 2as$$

# Motion under gravity

\* When motion takes place under the influence of gravity



$$\textcircled{1} \quad v = u + at \quad a = -g$$

$$v = u + -gt$$

$$v = u - gt \quad \checkmark$$

$$\textcircled{2} \quad s = ut + \frac{1}{2}at^2$$

$$s = ut + \frac{1}{2}(-g)t^2$$

$$s = ut - \frac{1}{2}gt^2 \quad \checkmark$$

$$\textcircled{3} \quad v^2 = u^2 + 2as \quad \xrightarrow{-g}$$

$$v^2 = u^2 + 2(-g)s$$

$$v^2 = u^2 - 2gs \quad \checkmark$$

**Thank  
You**